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Machine learning-based compressive strength prediction for concrete: An adaptive boosting approach

De-Cheng Feng [a,b,*], Zhen-Tao Liu [b], Xiao-Dan Wang [c], Yin Chen [c], Jia-Qi Chang [b], Dong-Fang Wei [b], Zhong-Ming Jiang [d,*]

[a] Key Laboratory of Concrete and Prestressed Concrete Structures of the Ministry of Education, Southeast University, Nanjing 211189, China
[b] School of Civil Engineering, Southeast University, Nanjing 211189, China
[c] College of Software Engineering, Southeast University, Nanjing 211189, China
[d] College of Engineering & Technology, Southwest University, Chongqing 400045, China

HIGHLIGHTS

- The AdaBoost algorithm is adopted to predict the compressive strength of concrete.
- 1030 sets of data is collected to train the model and reaches an accuracy of 98%.
- Different algorithms are compared to show the superior of the proposed model.
- Key factors and in the AdaBoost and influence of input variables are investigated.

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ABSTRACT

In this paper, an intelligent approach based on the machine learning technique is proposed for predicting the compressive strength of concrete. This approach employs the adaptive boosting algorithm to construct a strong learner by integrating several weak learners, which can find the mapping between the input data and output data. The weak learner whose predicting error is small will have a larger weight in the entire system, thus the overall accuracy of the strong learner will be enhanced. A total of 1030 sets of concrete compressive strength tests is collected to train and test the learners, in which the concrete mixture components (e.g., coarse/fine aggregates, cement, water, additive, etc.) and the curing time are set as the input data while the compressive strength value is set as the output data. The proposed approach is validated through a 10-fold cross validation method, and reaches an average accuracy of over 95% in sense of determination coefficient. In addition, a new dataset of 103 samples for concrete compressive strength is also adopted to demonstrate the generalization power of the proposed mode. The proposed approach is also compared to some other individual machine learning techniques that are already applied in this field, e.g., artificial neural network (ANN) and support vector machine (SVM), and shows superior advantages over these methods. Finally, the influence of some key factors in the adaptive boosting approach is also investigated, e.g., the amount of training data, the choice of weak learner, and the influence of the sensitivity and number of the input parameters. It is shown that using 80% of the total data for training can obtain acceptable prediction results and decision tree is the best choice for the weak learner in the boosting framework. Also, the importances of different input variables are obtained based on the sensitivity analysis results.

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1. Introduction

Concrete is the most widely used building material around the world due to its various advantages over other materials, e.g., integrity, durability, modularity, economy, etc. For better understanding the behavior of concrete-made structures under external

* Corresponding authors at: Key Laboratory of Concrete and Prestressed Concrete Structures of the Ministry of Education, Southeast University, Nanjing 211189, China (D.-C. Feng); College of Engineering & Technology, Southwest University, Chongqing 400045, China (Z.-M. Jiang).

E-mail addresses: dcfeng@seu.edu.cn (D.-C. Feng), jiangzm@swu.edu.cn (Z.-M. Jiang).

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loadings and developing corresponding design methodologies, it is of great importance to study the mechanical properties of concrete. Among the various indices of properties of concrete, compressive strength is the most fundamental one since it is directly related to the safety of the structures, and it is required in the performance determination of the structures during the whole life-cycle, from new structural design to old structural assessment. However, as it is known to all, concrete is made up of different components, e.g., coarse/fine aggregates, cement pastes, additional mixtures, etc., and these components are randomly distributed over the entire concrete matrix. Such a complicated system makes it a great challenge to accurately predict the compressive strength of concrete material [1,2].

Generally, the most direct way to obtain the compressive strength of concrete is through physical experiments. Usually the cubic or cylinder specimens were prepared according to certain designed mixture ratio and then cured for a required time. After that by using the compressive test instrument the compressive strength can be obtained easily, see [3–5]. However, this approach is costly in both time and money, thus the working efficiency will be very low. Differing from the traditional experimental ways, also some empirical regression methods are proposed to predict the concrete compressive strength with the given designed mixture ratio of different components in concrete, see [6–8]. Unfortunately, the concrete mixture and compressive strength shows a strongly nonlinear relation, thus it is difficult to derive an accurate regression expression for this problem. The third way to capture the concrete behavior is numerical simulation, see [9,10], etc. However, as mentioned before, the coupling of randomness and nonlinearity makes it different to accurately reproduce the concrete behavior [1,11].

On the other hand, with the development of artificial intelligence (AI) in recent years, it is a trend to use machine learning (ML) techniques to predict the concrete compressive strength. ML is a branch of AI, and can be used for several objectives, e.g., classification, regression, clustering, etc. Predicting concrete compressive strength is just one application of the regression function of ML. Compared with other traditional regression method, ML adopts certain algorithms that can learn from the input data themselves and gives highly accurate results for the output data, which shows an obvious advantage over the traditional regression methods [12].

Up to date, several ML algorithms are used for concrete compressive strength prediction, among which the most preferred ones are artificial neural network (ANN) and support vector machine (SVM). To name a few, Siddique et al. [13] used ANN to predict the compressive strength of self-compacting concrete containing bottom ash. Uysal and Tanyildizi [14] adopted ANN to estimation the compressive strength of self compacting concrete after exposed to high temperature. Dantas et al. [15] and Duan [16] applied ANN for strength prediction of concrete containing construction waste or recycled aggregate concrete, respectively. Chou et al. [17,18] investigated several ML techniques in predicting the concrete compressive strength, including both ANN and SVM. Aiyer et al. [19] introduced an advanced version of SVM, namely, least square SVM, to this field. Motamedi et al. [20] extended the SVM-based prediction method to a more complex problem, i.e., unconfined compressive strength of cockle shell "cement" sand mixtures. Pham et al. [21] further improved the least square SVM using the metaheuristic optimization, and used it to predict compressive strength of high-performance concrete. Omran et al. [22] compared the accuracy of different data mining techniques for predicting the compressive strength of environmentally friendly concrete. Chithra et al. [23] studied application of ANN in predicting compressive strength of concrete containing nano silica and copper slag. Naderpour [24] also used ANN to predict the strength of environmentally friendly concrete. Ashrafian [25] used heuristic regression methods in prediction of both strength and ultrasonic pulse velocity for fiber reinforced concrete. Zhang et al. [26] used random forest (RF) to predict the compressive strength and discussed the importance of each input variables.

However, most of the above-mentioned algorithms are individual learning ones, while there is another family of ensemble learning algorithms [27], which is more accurate, robust and powerful. The basic idea of ensemble learning models is to train several weak learners using the training data, and then integrate the weak learners to construct a strong learner. The weak learners are actually based on the individual learning algorithms, e.g., ANN or SVM. Therefore, the predicting accuracy and robustness of ensemble learning models (namely, the strong learner) will be obviously improved. There are three major groups of algorithms for ensemble learning, i.e., bagging, boosting and stacking, whose differences can be found in the review paper [28].

Among the various ensemble learning algorithms, adaptive boosting (AdaBoost) [29] is the most typical one. In this paper, we proposed a new intelligent approach based on the AdaBoost algorithm for predicting the compressive strength of concrete. A total of 1030 sets of concrete compressive strength tests are collected, and the mixture contents and curing time are set as the input data while the compressive strength is set as the output data. Then the fundamental mathematical background of AdaBoost and the corresponding implementation are introduced. By training the AdaBoost-based model using the collected data, a strong learner which can be used to predict the compressive strength value is obtained. A 10-fold cross validation method is used to validate the developed model, and a new data set including 103 samples is also used to show the generalization of the model. The performance of the AdaBoost model is also compared to those previous individual ML models, e.g., ANN-based and SVM-based, indicating the superiority of the ensemble learning method. Finally, the influences of some key factors in the AdaBoost approach are also discussed, e.g., the amount of the training data, the type of the weak learner and the sensitivity and number of the input variables.

2. Experimental database

To build up the prediction model, a large amount of the experimental data on concrete compressive strength is required. To this end, we have collected 1030 concrete compressive strength tests in literatures. In fact, the data set was originally collected by Yeh [30,31], and was already used for training and testing individual machine learning algorithms, see [18]. The tested concrete has eight ingredients, i.e., the ordinary Portland cement, water, coarse aggregate, fine aggregate, super-plasticizer, blast-furnace slag and fly ash, and it was cured under normal condition for some time before test. Then the compressive strength of the concrete was obtained through a typical compressive test procedure of the cylinder specimens with height 150 mm. Obviously, there are 9 parameters in total in the experimental data sets. The name, unit, min/max values, mean value and standard derivation (SD) of the experimental parameters are listed in Table 1. Moreover, the statistical distribution of the involved parameters are plotted in Fig. 1, which can help us have a direct observation of the parameters.

Before starting the learning procedure, the input and output variables should be determined at first. Evidently, the ingredients and the curing time both have an influence on the finial compressive strength of the concrete. Therefore, in total 8 input variables ($\boldsymbol{X} = \{X_1, X_2, \ldots, X_8\}$) and 1 output variable (Y) are considered in this study, and are shown in Table 1.

Table 1
Parameters in the compressive strength tests.

Parameter	Unit	Minimum	Maximum	Mean	SD	Type
X_1: Cement	kg/m^3	102.00	540.00	281.17	104.51	Input
X_2: Water	kg/m^3	121.75	247.00	181.57	21.36	Input
X_3: Coarse aggregate	kg/m^3	801	1145.0	972.92	77.75	Input
X_4: Fine aggregate	kg/m^3	594.00	992.60	773.58	80.18	Input
X_5: Super-plasticizer	kg/m^3	0.00	32.20	6.20	5.97	Input
X_6: Blast-furnace slag	kg/m^3	0.00	359.40	73.90	86.28	Input
X_7: Fly ash	kg/m^3	0.00	200.10	54.19	64.00	Input
X_8: Curing time	days	1.00	365.00	45.66	63.17	Input
Y: Compressive strength	MPa	2.33	82.60	35.82	16.71	Output

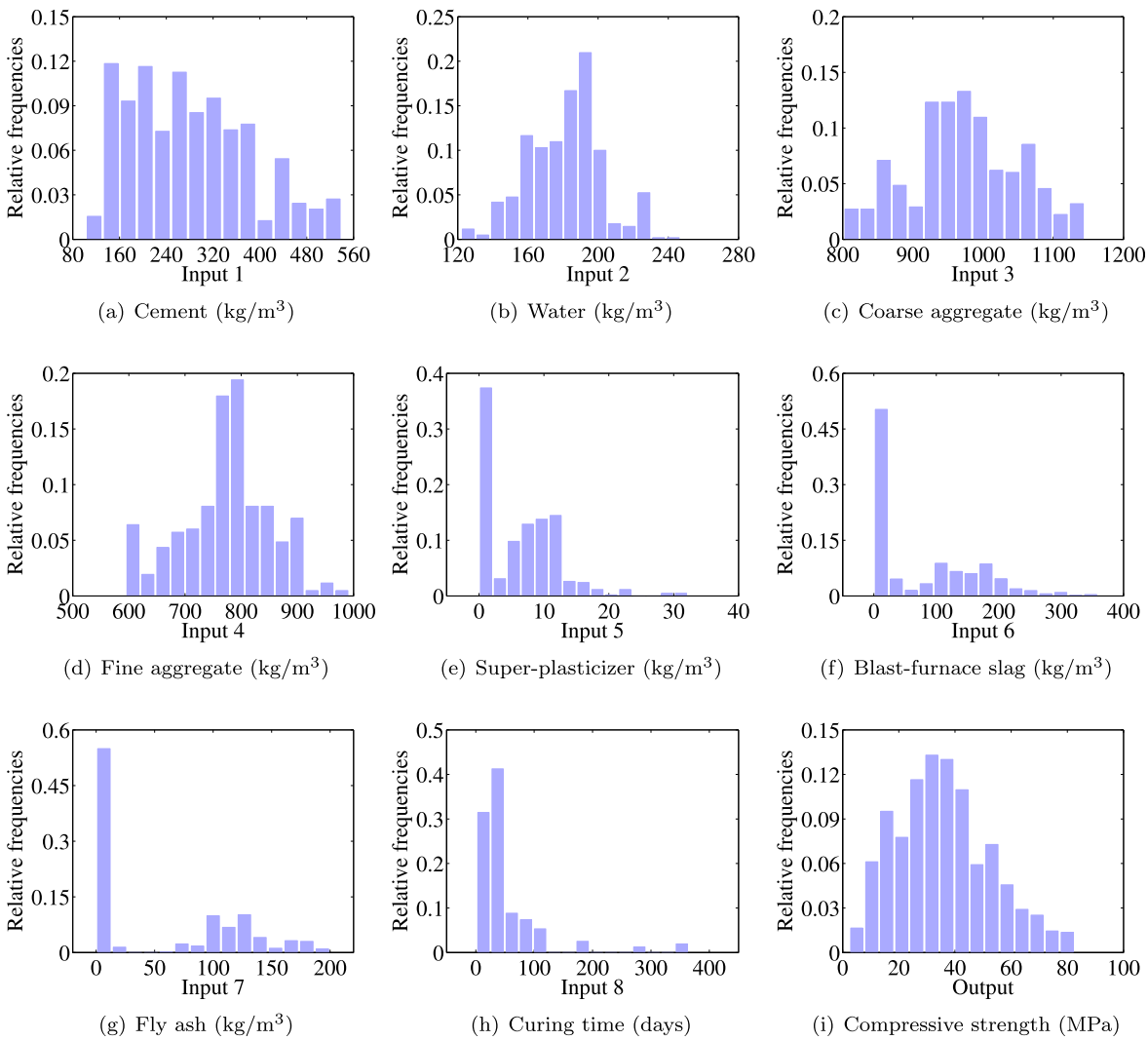

Fig. 1. Statistical distributions of the input/output variables.

3. Adaptive boosting algorithm

As mentioned before, AdaBoost algorithm is the most typical and widely used algorithm of ensemble learning method, or more specifically, of the boosting family of ensemble learning. The characteristic of AdaBoost is to use the initial training data to generate a weak learner, then adjust the distribution of the training data according to the predicting performance for the next round weak learner training. Note that the training samples with low predicting accuracy in the previous step will get more attention in the next step. Finally the weak learners are integrated together with different weights to a strong learner. The detailed mathematical background and implementation process will be presented as follows.

3.1. Mathematical background

The AdaBoost can be used for both classification and regression. In this paper, we use the AdaBoost regression to predict the concrete compressive strength. Considering a generally regression problem, the training data set $\boldsymbol{\Theta}$ can be denoted as

$$\boldsymbol{\Theta} = \{(\boldsymbol{X}_1, Y_1), (\boldsymbol{X}_2, Y_2), \ldots, (\boldsymbol{X}_m, Y_m)\} \tag{1}$$

where $(\boldsymbol{X}_i, Y_i)(i = 1, \ldots, m)$ is the $i-th$ sample in the training data set and m is the total number of the samples; $\boldsymbol{X}_i$ is the input data vector; Y_i is the output data value.

Then it can be used to train a base learner (or weak learner) $G(\boldsymbol{X})$ using some specific learning algorithms, and the relative predicting error e_i on each sample can be written as

$$e_i = L(Y_i, G(\boldsymbol{X}_i)) \tag{2}$$

where $L(\cdot)$ is the loss function, which usually has three options, i.e., linear loss, square loss and exponential loss. For simplicity, the linear loss function is adopted, namely,

$$e_i = \frac{|Y_i - G(\boldsymbol{X}_i)|}{E} \tag{3}$$

in which $E = \max|Y_i - G(\boldsymbol{X}_i)|$ is the maximum absolute predicting error of all the samples.

Obviously, the performance of only one weak learner will be very poor, so the purpose of the AdaBoost is to sequentially produce a series of weak learners $G_k(\boldsymbol{X}), k = 1, 2, \ldots, N$, and then combine them to construct a strong learner $H(\boldsymbol{X})$ by some combination strategy. For regression problem, the combination strategy is

$$H(\boldsymbol{X}) = v\sum_{k=1}^{N}\left(\ln\frac{1}{\alpha_k}\right)g(\boldsymbol{X}) \tag{4}$$

where α_k is the weight of the weak learner $G_k(\boldsymbol{X}); g(\boldsymbol{X})$ is the median of the all the $\alpha_k G_k(\boldsymbol{X}), k = 1, 2, \ldots, N; v \in (0, 1]$ is the regularization factor (or called as learning rate) which is used to avoid overfitting issue.

The weak learner $G_k(\boldsymbol{X})$ and its weight α_k is produced using the modified versions of the original training data, which is achieved by a so-called re-weighting approach, namely, adjusting the distribution weights of each sample according to the predicting error by previous weak learner $G_{k-1}(\boldsymbol{X})$. The mis-predicted samples will have their weight increased such that they will be more concentrated on in the next training process. Evidently, this is an iterative process. For iteration $k = 1, 2, \ldots, N$, the weak learner is $G_k(\boldsymbol{X})$ and the relative predicting error e_{ki} is calculated according to Eq. (3), thus the total error ratio e_k of this step is expressed as

$$e_k = \sum_{i=1}^{m} e_{ki} \tag{5}$$

Then the weight of the weak learner is

$$\alpha_k = \frac{e_k}{1 - e_k} \tag{6}$$

and the distribution weight $w_{k+1,i}$ of each sample for next step training is updated as

$$w_{k+1,i} = \frac{w_{k,i}\alpha_k^{1-e_{ki}}}{\sum_{i=1}^{m} w_{k,i}\alpha_k^{1-e_{ki}}} \tag{7}$$

Note that there are two kinds of weight in the above derivations, i.e., $w_{k,i}$ and α_k. The first one ($w_{k,i}$) is in the sense of training data samples, which is used to indicate that the mis-predicted samples will have their weights increased then they can be better learned in the next step. The second one (α_k) is in the sense of the weak learners, which is used to indicate that the more accurate weak learner will have a large influence on the finally results.

It can be also be seen form Eq. (4) that the AdaBoost is actually providing a powerful framework rather than a specific learning algorithm, since the detailed form of the weak learner $G(\boldsymbol{X})$ is not given. Theoretically, any kind of the machine learning regression algorithms can be used as the weak learner in AdaBoost. The widely adopted ones are Decision Tree (DT) regression [32] or ANN, here we use the DT type since it was successfully applied in several fields. However, the detailed theoretical background of DT will not be presented here since it can be easily found elsewhere and is out the scope of this paper.

3.2. Implementation process

The implementation of AdaBoost can be performed easily. In general it has four stages: (1) collection of the experimental data; (2) generation of the strong learner; (3) test or validation of the learner; (4) application of the learner to the engineering problems. Obviously, the second stage is the core of AdaBoost. As discussed above, it actually involves two levels, i.e., a framework to integrate the weak learners to a strong one and a regression learning algorithm to produce the weak learner using the training data. The DT algorithm, or more specifically, the classification and regression tree (CART) [33], is used to generate the weak learner, and the weak learners are integrated through the median of the weighted weak learners. The flowchart for this procedure is given in Fig. 2.

The parameters for the AdaBoost also includes two levels, one level is for the AdaBoost framework, and the other is for the algorithm of the weak learner, i.e., the CART. The framework has only two parameters, the number (or maximum iteration number) N of the weak learners and the regularization factor (or learning rate) v. The CART is relatively more complicated and has several parameters. For the concrete compressive strength prediction problem whose features are not large, the key parameters are: the maxi-

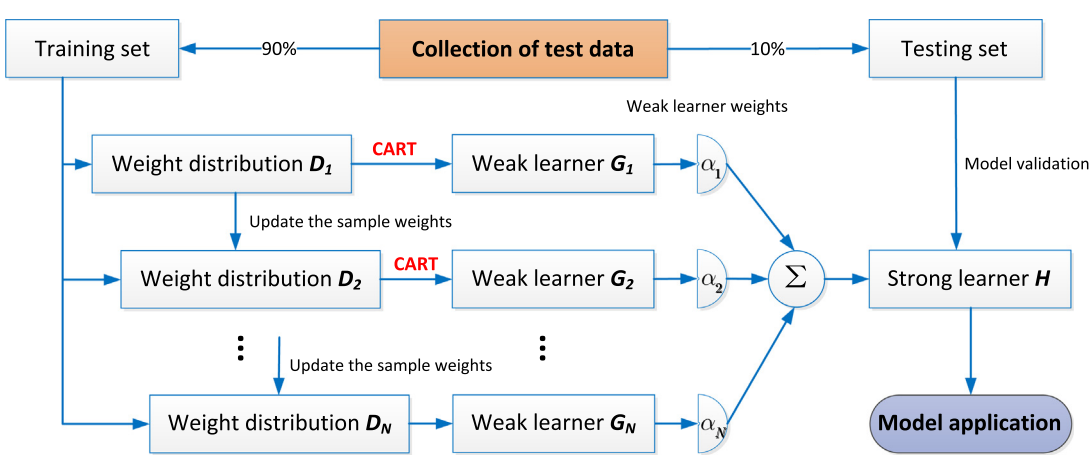

Fig. 2. Flowchart for implementation of the AdaBoost.

Table 2
Algorithm settings for AdaBoost.

Level	Parameter	Setting
Framework	Maximum iteration number	200
	Learning rate	0.2
Weak learner	Maximum depth blow root	50
	Minimum samples for split	5
	Minimum samples of leaf node	2
	Minimum impurity	10^{-4}

mum depth blow root, the minimum samples for split, the minimum samples of leaf node and the minimum impurity for further split. To achieve the best model performance, here an optimization strategy is employed to determine the above mentioned parameters. Initially, the value ranges of these parameters are determined according to previous researches in other subjects and literature, e.g., [18,34]. Then a grid search method is used to find the specific values, say, define a parameter grid from the initial value ranges and then iteratively train and test the model using cross-validation to find the parameter set with best performance. Finally, the optimized values for the above parameters are listed in Table 2.

4. Model validation

To build up the AdaBoost predictive model, the collected experimental data was split into two parts, i.e., the training set and the testing set. The training set is used to produced the final strong learner, and the testing set is used to show the accuracy of the model in predicting the compressive strength. Usually the percentage ratio of the two parts in total experimental data set is 90%–10% (9–1), while other ratios such as 80%–20% (8–2) can also be used. The results by the proposed method are discussed in this section and the influence of this ratio will be discussed later.

4.1. Single 9–1 validation results

Firstly, the single 9–1 results, i.e., randomly selected 90% data used for training and the remaining 10% data used for testing, are demonstrated here. Fig. 3 shows the relation between the predicted compressive strength values and the tested compressive strength values for the training set and testing set, respectively. Obviously, for both the two sets it displays a linear relation between the predicted and tested values. Especially, for the training set, the scatter points are nearly coincided with the ideal black lines in the figures (which means predicted value equals tested value), while for the testing set, there exists a little dispersion.

To better evaluate the performance of the predicting model by the AdaBoost algorithm, four different indicators are introduced, which are respectively defined as

-Coefficient of determination R-Squared (R^2)

$$R^2 = 1 - \frac{\sum_{i=1}^{m}(P_i - T_i)^2}{\sum_{i=1}^{m}(T_i - \overline{T})^2} \tag{8}$$

-Root Mean Squared Error (RMSE)

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{m}(P_i - T_i)^2}{m}} \tag{9}$$

-Mean Absolute Percentage Error (MAPE)

$$\mathrm{MAPE} = \frac{100\%}{m}\sum_{i=1}^{m}\left|\frac{P_i - T_i}{T_i}\right| \tag{10}$$

-Mean Absolute Error (MAE)

$$\mathrm{MAE} = \frac{\sum_{i=1}^{m}|P_i - T_i|}{m} \tag{11}$$

where P_i and T_i are the **P**redicted and **T**ested values, respectively; $\overline{T}$ the mean value of all the tested values; m is the total number of the samples in the data set.

Obviously, the measure R^2 shows the extent of the linear correlation between the predicted and tested values, the closer R^2 is to 1, the better the performance of the model. RMSE indicates the deviation between the predicted values and the tested values. MAE reflects the actual situation of the prediction error. MAPE, in addition to MAE, reflects both the prediction error and the ratio of the error to the tested value.

The computed four measures for the 9–1 results by the AdaBoost model is listed in Table 3, where the results for the training set and testing set are both given. Evidently, the model shows a very good performance in predicting the compressive values. For the training data set, $R^2 = 0.997$ which means the predicted value almost the same as the tested value; MAE is only 1.52 MPa and MAPE is only 5.54%. All these results indicate that the AdaBoost model has a very strong learning ability from the training data. For the testing data, the $R^2 = 0.982$ is still very close to 1, and the MAPE is relatively high, which is 6.78%. However, the error ratio is still very low in the viewpoint of engineering practice. In summary, the AdaBoost model produces compressive strength with very high accuracy.

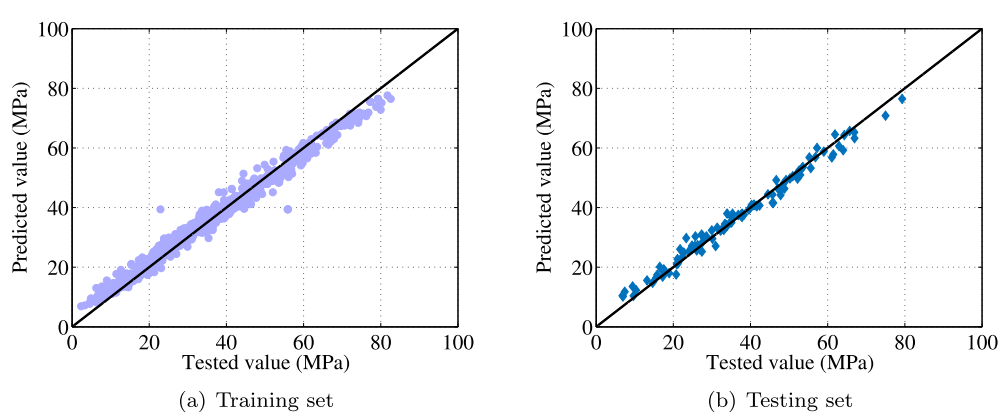

Fig. 3. Relation between tested and predicted compressive strength.

Table 3
Performance analysis of 9–1 results.

Data set	Performance measures			
	R^2	RMSE (MPa)	MAPE	MAE (MPa)
9–1 Train	0.997	1.52	5.54%	1.25
9–1 Test	0.982	2.20	6.78%	1.64

4.2. 10-fold cross validation results

To further demonstrate the performance of the AdaBoost model, the 10-fold cross validation method is employed. The method is developed to minimize the bias associated with randomly sampling of the training data set. It divides the experimental data samples equally into 10 subsets, and then uses 9 subsets to build up the strong learner while the left 1 subsets to validate the model. After successively repeating this operation for 10 times, we can obtain an average accuracy of the 10 times. It is reported that the 10-fold cross validation method can represent the generalization and reliability of the model performance [35].

Fig. 4 shows the four measures of each fold in the 10-fold cross validation. It can be seen that although the results of 10 folds show some fluctuations, it all keeps a high level of accuracy. For instance, the minimum value for R^2 is 0.890 while the maximum value is 0.978. The minimum MAPE is 7.96% while the maximum is 15.27%. Meanwhile, the detailed statistical information of the results is given in Table 4. The average of R^2 is 0.952 and the SD is 0.025, yielding a coefficient of variation (COV) of only 2.6%. The average value for RMSE is 4.856 MPa, relatively small compared with the mean compressive strength value 35.82 MPa. The MAE is 3.205 MPa and the MAPE is 11.39% in average, both indicate that the prediction error is small.

4.3. New data set results

To further demonstrate the generalization of the developed model, a new data set was employed as validation. The data set was provide by Yeh [36] and originally used for prediction of concrete slump. The data set includes 103 samples and each sample has 8 ingredients, say, cement, water, coarse aggregate, fine aggregate, super-plasticizer, blast-furnace slag, fly ash and 28-day compressive strength. It can be found that the only difference between this data set and the one used for training (shown in Table 1) is that the curing time is fixed to 28 days. So here we have 7 input variables (the 8-th input variable is defined as 28) and 1 output variable.

The predicted results for the new data set using the established AdaBoost model in the previous section are just shown in Fig. 5. As can be seen, the proposed model also yields a good prediction on the new data set. The relation between the predicted and tested values is very close to the ideal $y = x$ line. The computed quantitative performance measures are: R^2=0.94, RMSE = 1.93 MPa, MAE = 1.43 MPa and MAPE = 4.3%. All of these indicates the AdaBoost model really has a high accuracy in predicting the compressive strength of concrete, even for a totally new data set.

4.4. Time-dependent analysis results

With the established predictive model, it also enables us to obtain the time-dependent values of the compressive strength for concrete, as the curing time is also an input variable of the model. To demonstrate this feature, three groups of concrete mix proportions were selected. The information was listed in Table 5, and the curing time was set from 1 day to 365 days. Fig. 6 shows the obtained compressive strengths by AdaBoost model against time, where the scatter points are the test results in different time.

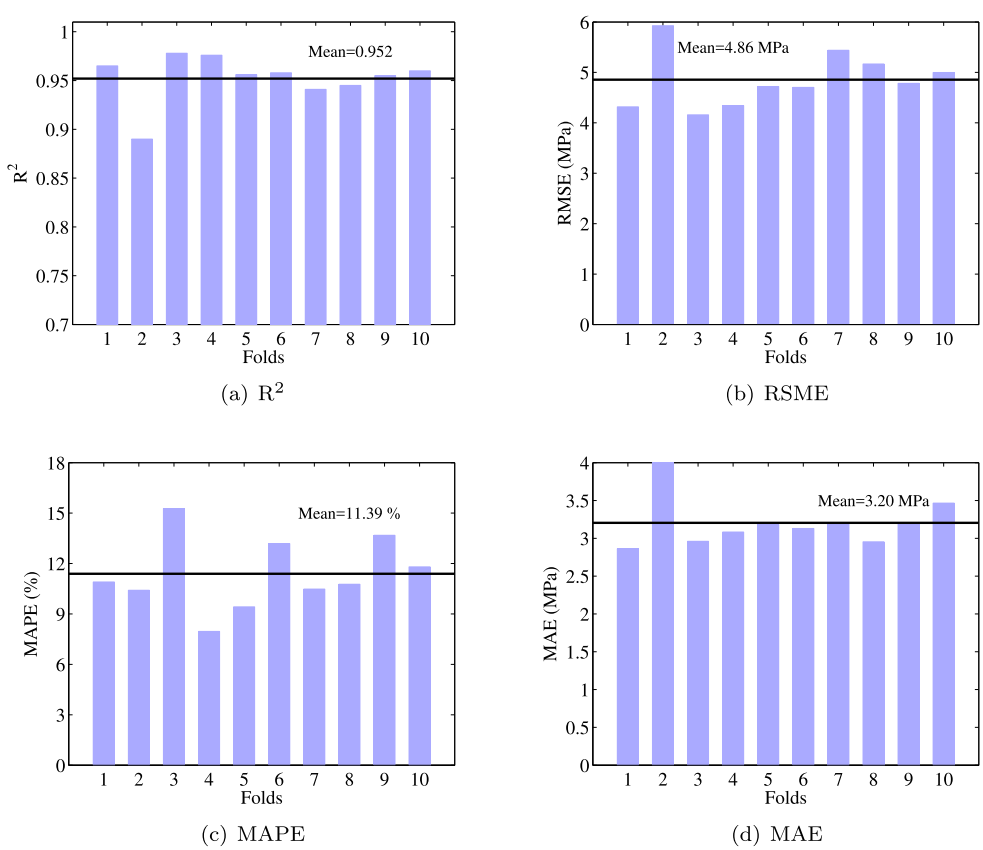

Fig. 4. Results for 10-fold cross validation.

Table 4
10-Fold cross validation results.

Folds	Performance measures			
	R^2	RMSE (MPa)	MAPE	MAE (MPa)
Fold 1	0.965	4.317	10.90%	2.862
Fold 2	0.890	5.927	10.41%	4.014
Fold 3	0.978	4.159	15.27%	2.960
Fold 4	0.976	4.343	7.96%	3.083
Fold 5	0.956	4.720	9.42%	3.180
Fold 6	0.958	4.704	13.19%	3.130
Fold 7	0.941	5.440	10.48%	3.204
Fold 8	0.945	5.168	10.77%	2.954
Fold 9	0.955	4.784	13.68%	3.200
Fold 10	0.960	4.999	11.80%	3.465
Average	0.952	4.856	11.39%	3.205
SD	0.025	0.547	2.15%	0.331

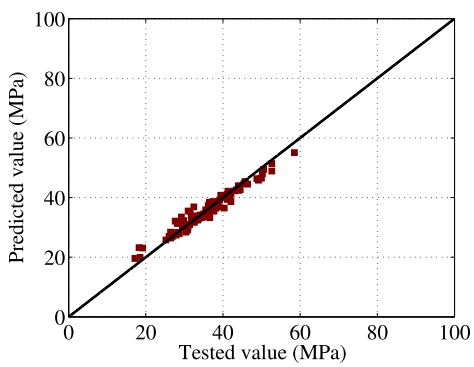

Fig. 5. Results for the new data set.

Obviously, the test data points are nearly all located at the time-varying curves. The curves of three groups have the similar tendency, i.e., the compressive strength will increase with the growing of time, and it increases rather fast before 60 days and then increases slowly after 60 days until to a steady value approximately after 120 days. The 7-day strength is around 50% of the final strength and 28-day strength is around 85% of the final strength. Obviously, this AdaBoost model can help us capturing a full picture of the concrete strength evolution against time without experiments, which can provide us an efficient tool to study other physical mechanisms, e.g., to calculate the resistance of a concrete member using the time-varying compressive strength.

5. Comparison with other machine learning techniques

To better exhibit the capacity of the AdaBoost algorithm, which is an ensemble learning method, the famous single learning methods that are widely adopted are also used in this paper, i.e., ANN and SVM. The determination of the model parameters are similar with that for AdaBoost, i.e., setting the initial values for the parameters and then using grid search to find the optimal values. The plots of the predicted versus tested values are shown in Fig. 7. Obviously, the results by AdaBoost display a more linear relation which means its predictions are more close to the tested ones.

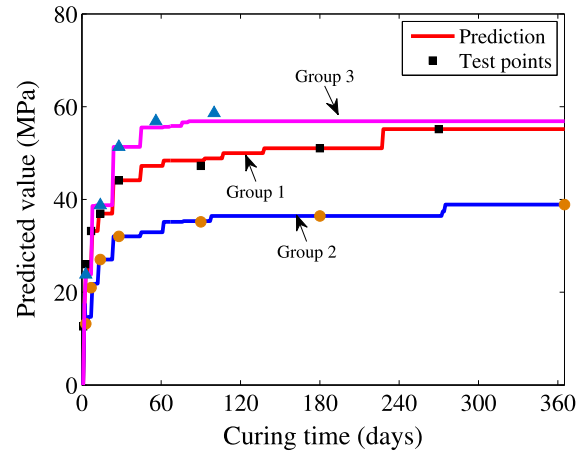

Fig. 6. Results for time-dependent prediction of concrete compressive strength.

The reason may be that either SVM or ANN is individual learning algorithm while AdaBoost is ensemble learning algorithm. As is known to all, ensemble learning integrates several weak learners generated by individual learning algorithms, and weak learner with good performance will got a higher weight while weak learner with bad performance will got a lower weight. Therefore, it can provide more accurate and robust predictions. The measures for the testing set by different algorithms are given in Table 6, which are similar to the results obtained by Chou et al. [17] and Erdal et al. [34]. The AdaBoost model shows an obvious superior over individual learning methods for all the four measures. The R^2 increase from 0.855 (SVM) to 0.982, while the MAPE decrease from 14.95% (SVM) to 6.78%. The improvement of the performance is dramatically.

Moreover, our results are also compare to some previous studies using individual learning algorithms, i.e., ANN, SVM, CART and linear regression (LR), just as shown in Table 7. Note that not all the four measures are provided by the references. Evidently, compared with the individual learning algorithms performed by previous researchers, whose highest R^2 is only 0.91, our model has a much better performance ($R^2 = 0.982$). Meanwhile, although some parameter tuning methods may be adopted to improve the performance of different models, like Chou et al. [18] whose minimum MAPE is 12.03%, it is still weaker than the AdaBoost model whose MAPE is only 6.78%.

In addition, the ensemble learning method is initially tried in some literatures, e.g., Erdal et al. [34] used gradient boosted ANN etc. and Chou et al. [18] used stacking method combining different individual learning algorithms. The best results reported in their works are compared to the AdaBoost model in Table 8. Similarly, their performance is still worse than AdaBoost.

6. Model performance analysis

6.1. Influence of training data set amount

As discussed before, the performance of the model also depends on the amount of data used for training. Therefore, in this part, we

Table 5
Concrete mix proportion for the time-dependent analysis.

Group	Mix proportion (kg/m^3)							Time (days)
	Cement	Slag	Fly ash	Water	Plasticizer	Coarse aggregate	Fine aggregate	
1	500.0	0.0	0.0	200.0	0.0	1125.0	613.0	1:365
2	339.0	0.0	0.0	197.0	0.0	968.0	781.0	1:365
3	275.1	0.0	121.4	159.5	9.9	1053.6	777.5	1:365

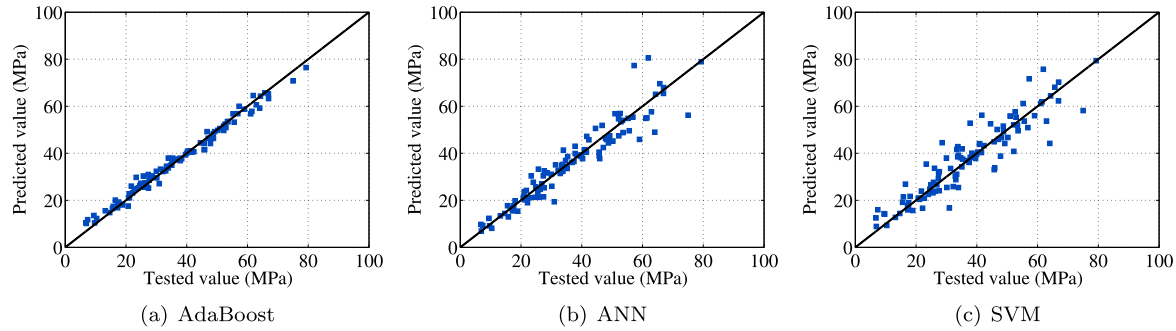

Fig. 7. Results for different machine learning techniques.

Table 6
Comparison with different machine learning techniques.

Algorithm	Performance measures			
	R^2	RMSE (MPa)	MAPE	MAE (MPa)
AdaBoost	0.982	2.20	6.78%	1.64
ANN	0.903	5.14	9.77%	3.41
SVM	0.855	6.28	14.95%	4.44

Table 7
Comparison with results for the same data in previous individual learning studies.

Reference	Algorithm	Performance measures			
		R^2	RMSE (MPa)	MAPE	MAE (MPa)
This paper	AdaBoost	0.982	2.20	6.78%	1.64
Chou et al. [17]	ANN	0.886	–	–	–
	SVM	0.910	–	–	–
Chou et al. [18]	ANN	–	7.95	20.84%	6.19
	SVM	–	5.59	12.03%	3.75
	CART	–	7.84	20.66%	5.86
	LR	–	10.11	29.89%	7.87
Erdal et al. [34]	ANN	0.909	–	–	–

Table 8
Comparison with results for the same data in previous ensemble learning studies.

Reference	Algorithm	Performance measures			
		R^2	RMSE (MPa)	MAPE	MAE (MPa)
This paper	AdaBoost	0.982	2.20	6.78%	1.64
Chou et al. [18]	Stacking with CART + SVM + LR	–	5.08	11.97%	3.52
Erdal et al. [34]	Gradient boosted ANN	0.953	–	–	–

try to investigate the influence of the training data set amount. Three cases are considered herein, i.e., the total data is divided for training and testing as 9–1, 8.5–1.5 and 8–2. The results are shown in Table 9. Obviously, all the three cases got very high accuracy, though it with still grow with the increase of the training data amount. For the training data set, no matter 9–1, 8.5–1.5 or 8–2, the performance is still very good. The R^2 for three cases are all close to 1.0, which means the relation between the predictions and test values is linear. The RMSE are around 1.52 MPa for 9–1 and 8.5–1.5, while it is 1.55 MPa for 8–2. This means that with the increase of the training data amount, the prediction deviation will decrease. Similarly, the MAPE will reduce to below 5.54% if we use more than 85% (875 samples) of the total data for training.

The conclusion is similar for that of the testing data set. If a larger amount of training data is imposed, both the prediction error and deviation will show a tendency of reducing. Specifically, the

Table 9
Different training data set amount results.

Data set	Performance measures			
	R^2	RMSE (MPa)	MAPE	MAE (MPa)
9–1 Train	0.997	1.52	5.54%	1.25
9–1 Test	0.982	2.20	6.78%	1.64
8.5–1.5 Train	0.997	1.53	5.54%	1.26
8.5–1.5 Test	0.981	2.25	7.89%	1.66
8–2 Train	0.996	1.55	5.74%	1.26
8–2 Test	0.979	2.26	7.78%	1.66

R^2 will grow form 0.979 to 0.982 when the training data increase from 80% to 90% of the whole set. The MAPE will decrease from 7.78% to 6.78%.

6.2. Influence of weak learner types

Another key factor in the AdaBoost approach is the choice of the weak learner. Theoretically, any type of the weak learner can be used in the framework of AdaBoost. Here we tried three widely adopted ones for comparisons, i.e., LR, ANN and SVM. The results are shown in Fig. 8. It can be seen from the figure that except for LR, other three weak learners all get very good results. The quantitative measures are listed in Table 10. For LR, the measures are R^2=0.619, RMSE = 10.19 MPa and MAPE = 31.55%, while for CART, ANN and SVM, the worst measures are R^2 = 0.960, RMSE = 3.30 MPa and MAPE = 8.58%. This is because LR assumes a linear relation between the input and output, which is actually nonlinear in reality. Consequently, though the AdaBoost approach was adopted to enhance the model performance, it still yields a poor result. By contrast, CART, ANN and SVM themselves can capture the nonlinear relation between the inputs and output already, so if the ensemble approach is used, the model performance can get increased. Also, as you can see in Table 10, selecting CART, ANN and SVM as the weak learner all has acceptable accuracy since these weak learners themselves can obtain the nonlinear relation between inputs and output to some extent.

6.3. Influence of input variable sensitivities

The developed AdaBoost model provides very high accuracy in predicting the compressive strength of concrete, however, it is a black box model which can capture the mapping between inputs and output but can not explain the detailed mechanism behind the model. Even an AdaBoost model with considerable complexity, it is not possible to find out the mechanism underlying inputs, weak learners, and weights with regards to the problem under study. Therefore, unlike traditional mechanical models, checking

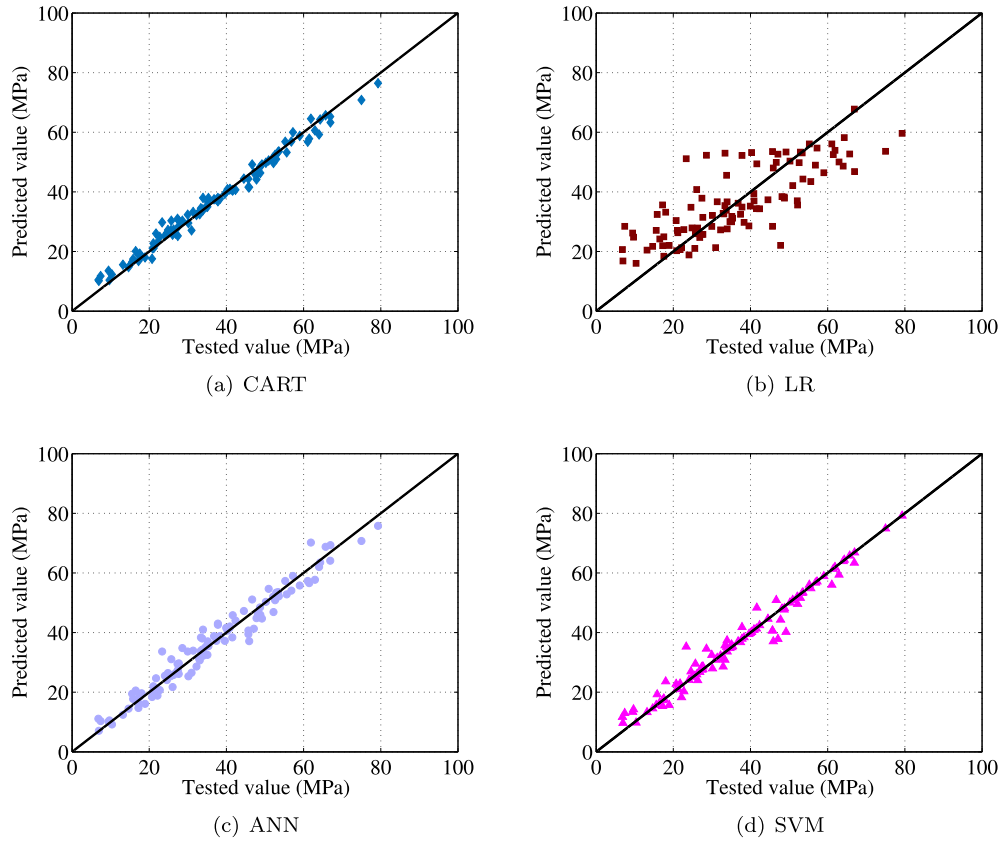

Fig. 8. Results for different weak learners.

Table 10
Performance measures for analysis of different weak learners.

Weak learner	Performance measures			
	R^2	RMSE (MPa)	MAPE	MAE (MPa)
CART	0.982	2.20	6.78%	1.64
LR	0.619	10.19	31.55%	7.95
ANN	0.960	3.30	8.58%	2.60
SVM	0.968	2.95	7.40%	1.73

the relationship between each explicative and dependent variable in general ML models is highly non-trivial.

However, though the model is a black box, we can also use it as a tool to study the underlying physical phenomena, e.g., to study the sensitivity of each input variable on the final compressive strength, thus we can got a sense about the importance of each ingredient. To achieve this goal, first we select some out-of-bag (OOB) samples, and randomly shuffle the values of a selected input while keeping the remaining input unchanged, then calculate a new output using the AdaBoost model. The difference between the accuracy of the two predictions are defined as the importance of the input variable. Fig. 9 shows the relative importance of each ingredients. It can be found that curing time is the most critical factor affecting the final compressive strength. It makes some sense since we can see from Fig. 6 that there is a dramatic grow of the compressive strength during 0 to around 80 days. Cement also has significant influence on the compressive strength, around 97% as curing time, as physically it cemented all other components together through hydration reaction. Similarly, water ranked third (57% as curing time) since it is used to ensure the hydration of cement and the fluidity of concrete. Coarse and fine aggregates have close contributions (25% as curing time) to the compressive

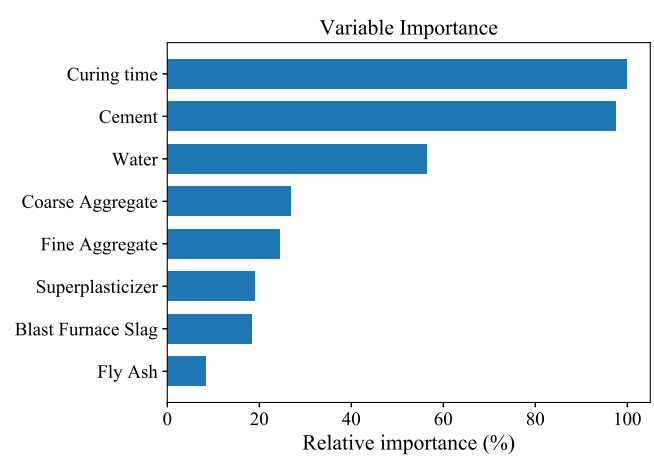

Fig. 9. Relative importance of different input variables.

strength, they actually form the main framework of concrete though they are cemented by cement and water. Admixtures like super-plasticizer, slag and fly ash have the least importance (10%–20% as curing time) since their functions are just to improve or adjust the performance of concrete or to meet the special requirements of concrete design. As discussed before, these sensitivities actually accord with the engineering practice and physical background.

6.4. Influence of input variable number

The influence of the sensitivity (or variation) of the input variable values are studied in the last part, here the influence of the

Table 11
Performance measures for analysis of different input variable combinations.

Combinations	Performance measures			
	R^2	RMSE (MPa)	MAPE	MAE (MPa)
1: $X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8$	0.982	2.20	6.78%	1.64
2: $X_1, X_2, X_3, X_4, X_5, X_6, X_7$	0.398	12.81	40.36%	10.05
3: $X_2, X_3, X_4, X_5, X_6, X_7, X_8$	0.865	6.07	16.42%	4.39
4: $X_1, X_2, X_4, X_6, X_7, X_8$	0.907	5.02	11.81%	3.46
5: X_2, X_5, X_6, X_7, X_8	0.753	8.20	21.89%	5.90
6: X_1, X_2, X_3, X_4	0.377	13.04	41.30%	10.43

input variable number on the finally prediction is analyzed. The purpose is to find out how the model performance changes if we missed some data (or inputs). Six combinations of inputs are considered, as listed in Table 11. Combination 1 is the raw data while combination 6 has only four inputs, i.e., cement, water, coarse aggregate and fine aggregate. Combination 2–5 gradually neglect different input variables to see the influence.

The results of predicted values vs. tested values by the six combinations are shown in Fig. 10, and the corresponding performance measures are also given in Table 11. For sure the results by combination 1 achieve the best scores since it provides the whole information as the inputs. The performance of combination 6 is lowest, whose measures are $R^2 = 0.377$, RMSE = 13.04 MPa and MAPE = 41.30%. It is interesting that the accuracy of the model will not simply increase with the increase of the input variable number, as indicated by the results of combinations 2–5. Though combination 2 has 7 input variables, it has the lowest accuracy (R^2=0.398, RMSE = 12.82 MPa and MAPE = 40.36%) since the most important variable curing time is not considered. As comparison, combination 3 also has 7 input variables (neglecting cement), its performance is much better than combination 2, whose R^2=0.865, RMSE = 6.07 MPa and MAPE = 16.42%. Combination 5 has only 5 input variables but it includes curing time, so its performance is better than combination 2 and 6. Combination 4 got the best performance (except combination 1) though it has only 6 input variables. In summary, curing time and cement is the most important input variables which should be considered in the model to get a high accuracy prediction. Neglecting curing time will lead to a dramatically drop of the accuracy (R^2 from 0.982 to 0.398), while neglecting cement will lead to a relatively small reduction of the accuracy (R^2 from 0.982 to 0.865). Neglecting one or even two other components will cause a even smaller drop (R^2 from 0.982 to 0.907).

7. Conclusions and discussions

In this paper, an ensemble learning method, AdaBoost algorithm, is adopted for the compressive strength prediction of concrete materials. 1030 sets of concrete compressive tests are collected and the test variables are divided into 8 inputs (e.g., coarse/fine aggregates, cement, water, additive, etc.) and 1 output (compressive strength value). The total data set is further split into training set and testing set. The model is generated by the AdaBoost from training set and then evaluated by the testing set. Based on the results, the following conclusions can be drawn:

- The AdaBoost model can accurately and efficiently predict the compressive strength of concrete with given input variables. The model is just like a black-box and the users do not need to know the actual mechanism, which makes it simple and convenient for application. The model can also capture the time-varying compression strength of concrete at different curing days.
- The average R^2 of the 10-fold cross validation may reach 0.952 and the average MAPE may reach 11.39%, which means the prediction error is very low. Meanwhile, the average RMSE is 4.856 MPa, indicating that the model also has a low prediction derivation. A new data set is also used to demonstrate the generalization of the model and the results indicate the model retains a high accuracy, whose $R^2 = 0.94$, RMSE = 1.93 MPa, MAE = 1.43 MPa and MAPE = 4.3%.
- The AdaBoost model is also compared to famous individual learning algorithms like ANN and SVM, and also demonstrates a obvious superior to these models. Meanwhile, the individual/ensemble learning algorithms that were applied in predict-

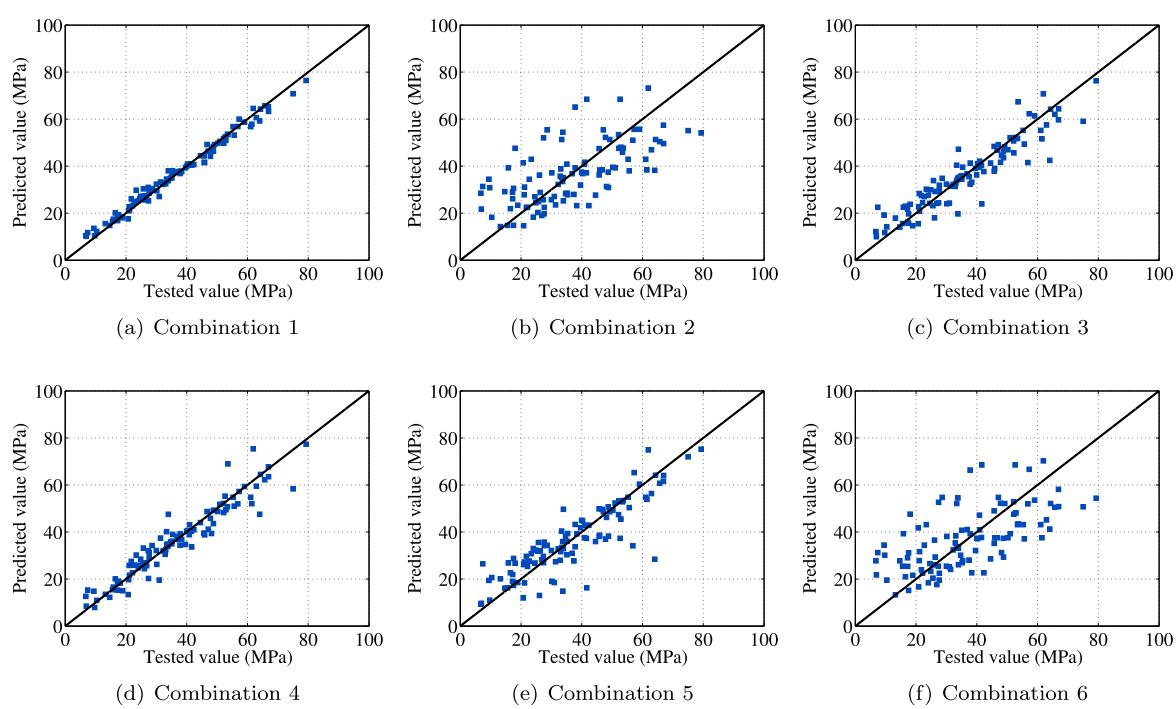

Fig. 10. Influence of number of input variables.

ing compressive strength in literatures are also adopted for comparison of the proposed AdaBoost model. Even the best results reported in the literatures are worse than the AdaBoost model.

- Several key factors in the AdaBoost model is also discussed, e.g., the training data set amount, the weak learner type, the sensitivity and number of input variables. It is found that using 80% of the whole data set can obtain the acceptable results. Selecting weak learners that can capture the nonlinear relation between the inputs and output (e.g., CART, ANN and SVM) can all obtain high accuracy of the final strong learner, while CART can achieve the best performance. Curing time, cement and water are the most important input variables and have very strong effect on the final predictions, which should not be neglected.

With the AdaBoost model, we can efficiently obtain the concrete compressive strength of different mix proportion at different time. Consequently, it can be used to check whether the designed mix proportion can satisfy the target strength requirement. Or it can be used to see the compressive strength growing rate at different age, helping engineers perform approval analysis during construction stage, safety analysis during service stage, etc. Moreover, the model can also be treated as a baseline model which can be extended to multi-output models which can predict several concrete properties at one time, e.g., strength, slumps and so on. Therefore, the model can be adopted for the optimization of the mix proportion for given strength and slumps restrains, namely, for development of new mix proportion of concrete with enhanced properties. These aspects will be studied based on the work presented in this paper in the near future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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